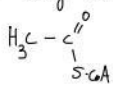


Cycle de l'Acide Citrique (Cycle de Krebs) = Cycle TCA

But: Oxydation complète Acétyl-CoA → Dérivés Carbone



$\rightarrow \text{NADH}$
 $\rightarrow \text{FADH}_2$

Energie réduite (eH apportés des e⁻)

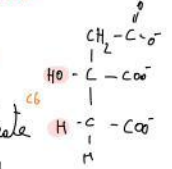
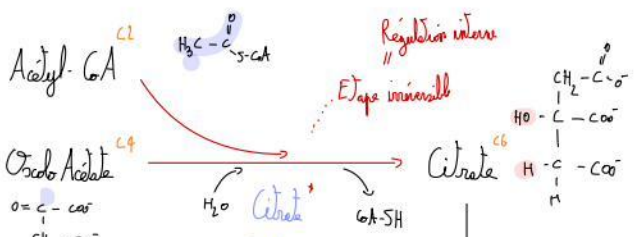
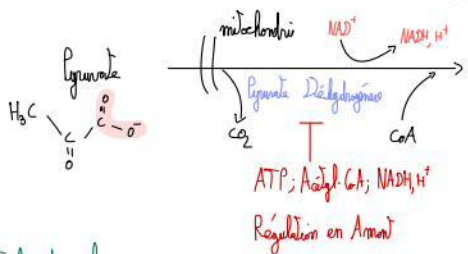
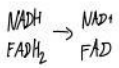
$\text{NAD}^+ \rightarrow \text{NADH}$

Leu: la matrice mitochondriale

Δ Enzyme Succinate déshydrogénase
 $\rightarrow$ accès à la mt interne
 $\hookrightarrow$ Complexe II chaîne respiratoire

E. Déshydrogénase = E. oxydante

Dépend indirectement de l'O₂ → chaîne respiratoire réoxygène

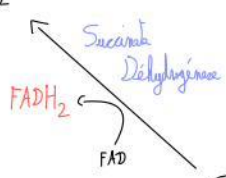
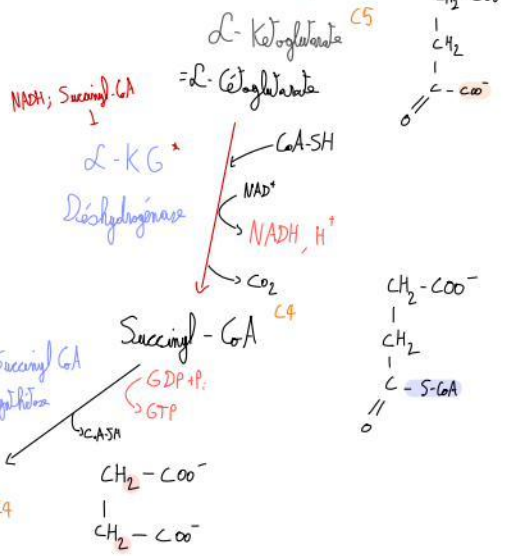
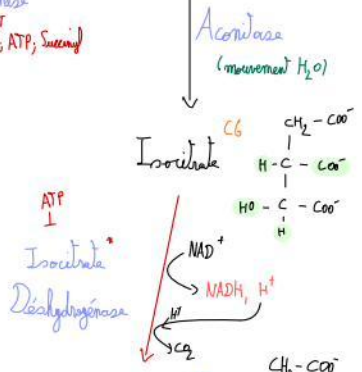
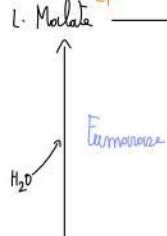
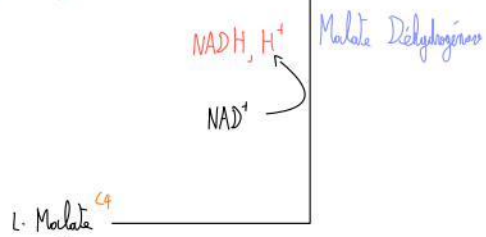
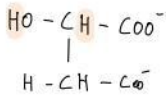


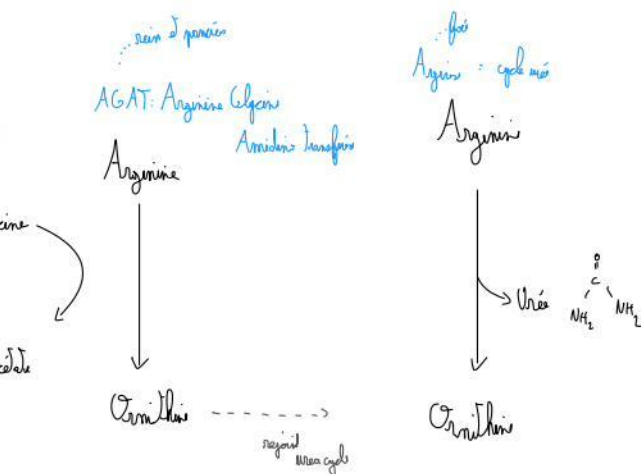
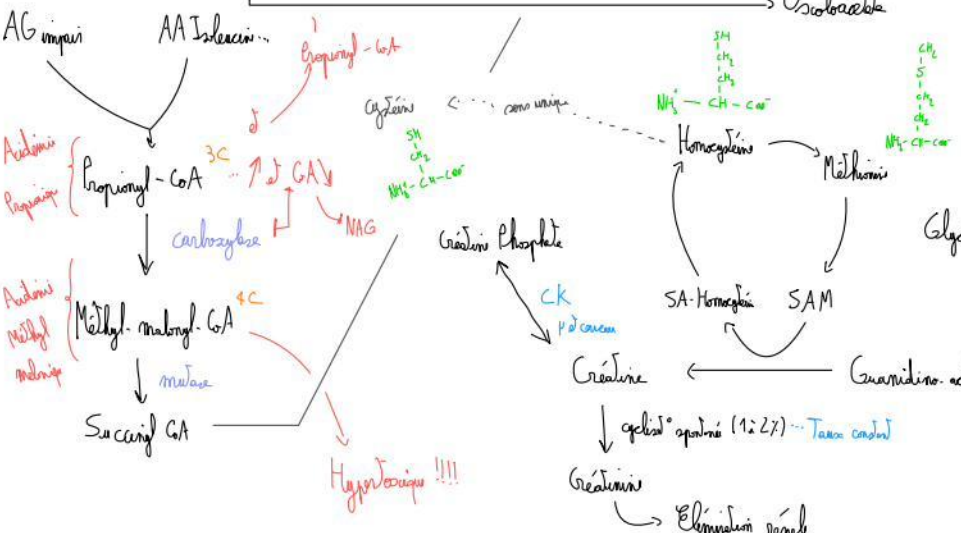
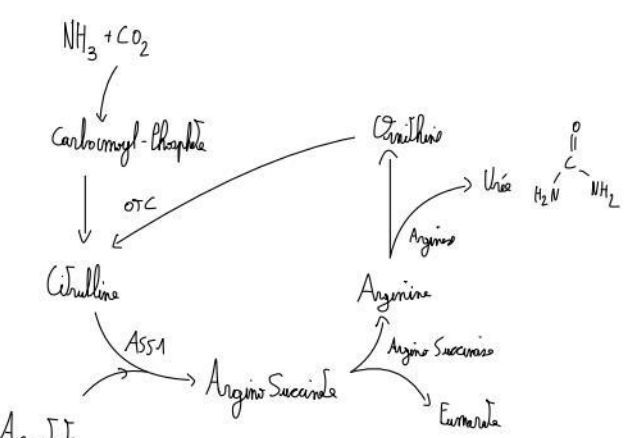
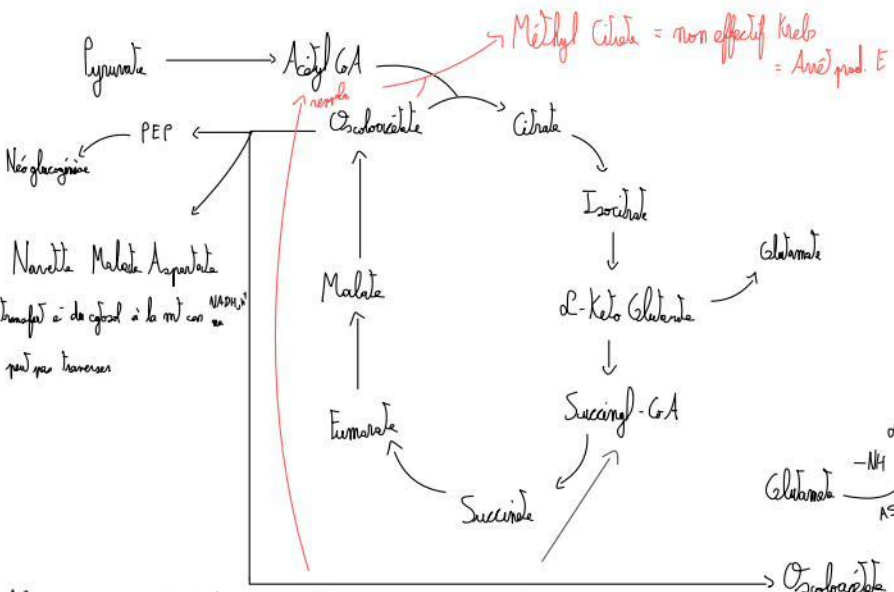
Bilan du cycle:

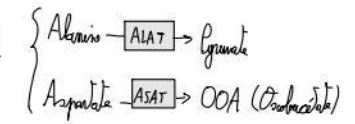
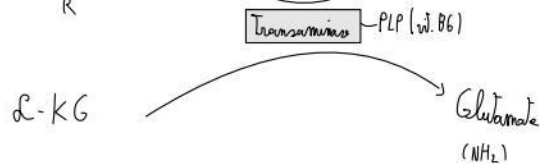
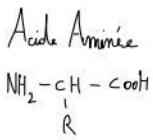
$3\text{NADH} \rightarrow 3 \times 3\text{ATP}$
 $1\text{FADH}_2 \rightarrow 2\text{ATP}$
 $1\text{GTP} \rightarrow 1\text{ATP}$
 2CO_2

12 ATP

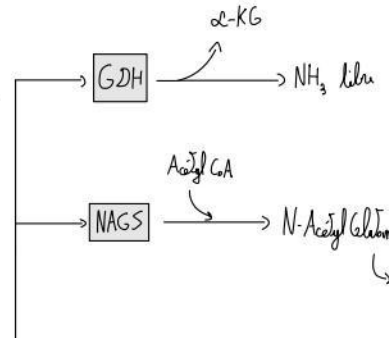
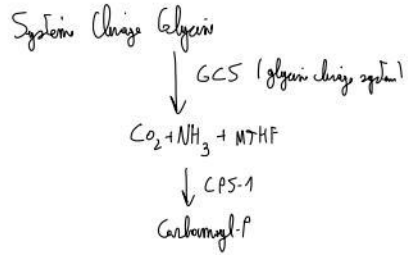
15 ATP avec la PDH



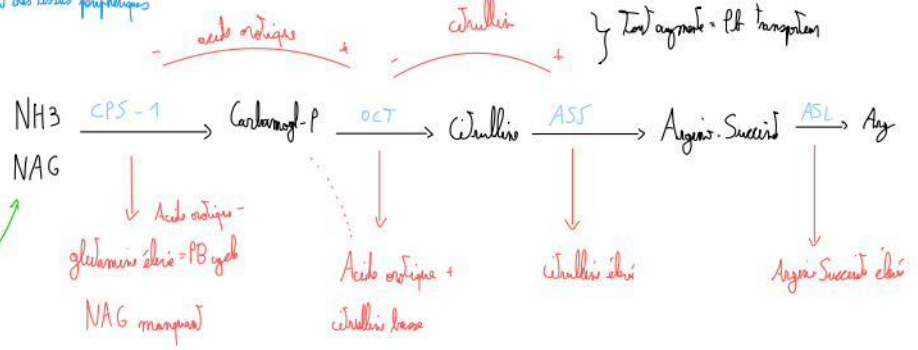
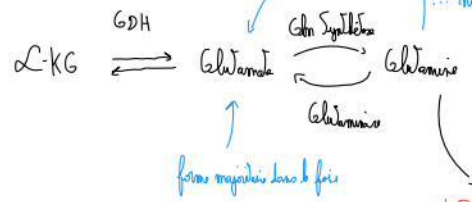
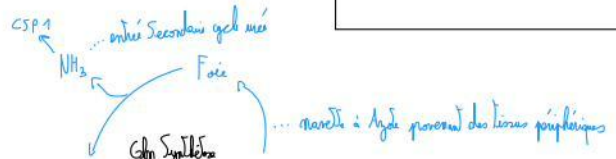
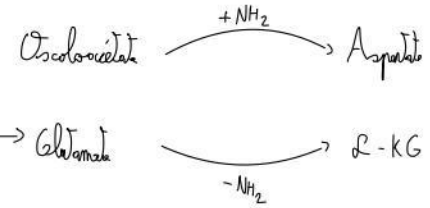
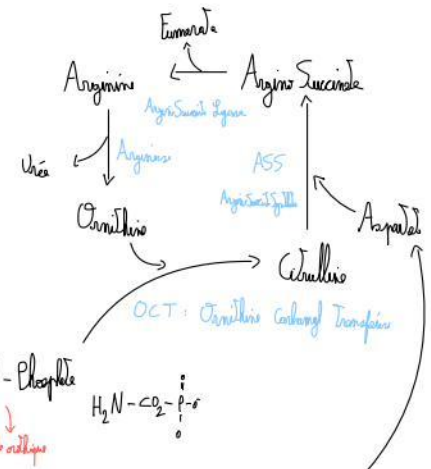
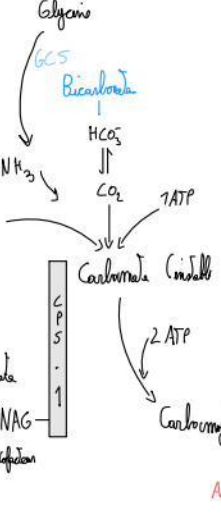




α-KG



voie métabolique glycine



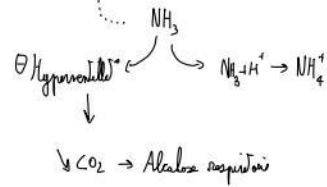
! Oedème cérébral!

"Carbapène" médicamenteux

^{glutamine élevée}
Hyperammonémie Primaire = maladie génétique $\rightarrow$ cause un UCD (Urea Cycle Disorder) $\downarrow$ NH_3^+ $\nearrow$ Alcalose

• Urogon = déficit en OCT

lié à l'X (surpression chez les filles)



Encephalite Hépatique $\rightarrow$ Trépanation au Zovirax $\rightarrow$ Finir = dégradé ses propres protéines $\rightarrow$ Vaisif Ammoniaque sont !

Dégradation Protéique = Risque Hyperammonémie (sport / fièvre / chimio / stress)

Médecines $\rightarrow$ Valproate (anticonvulsif)

^{glutamine basse}
Hyperammonémie secondaire = à glutamine basse

• enfant = canal Acarite

dij $\rightarrow$ urée général $\rightarrow$ cycle uré
^{Hyperammonémie}

• Krebs : $\downarrow$ 2-KG ; Acide Propionique